

Fossil palynomorph detection and classification

One joint model,
or a detector plus
a specialist classifier?

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A vast collection, examined one specimen at a time

~50 million

years into the past

~70,000

slides in the broader collection

Pollen helps reconstruct vegetation and warm climates.

The challenge is scaling expert identification.

Important features are small—and not always in focus



Subtle morphology



Staining differences



Debris and overlap

Multiple focal depths capture different parts of each specimen.

Build on detection; test how best to add identification

01 Whole-slide detection

Shaikh et al.: tiling, focal compression, and detection

02 Separate crop classification

Punyasena et al. and Martinsen et al.: two-stage precedents

03 Our comparison

Joint RF-DETR versus RF-DETR + Swin-Tiny

83 annotated slides; uneven category coverage

11,275

named annotations overall

2,290

initial target annotations

32

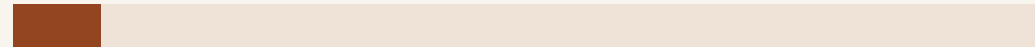
candidate target categories

Bisaccate



316

Cupuliferoipollenites sp.



27

Training options: undersampling, balanced batches, or weighted loss.

Turn multifocal slides into traceable model inputs

01

Define source groups

Separate slides and related samples before training.

02

Prepare image tiles

Combine focal information; read overlapping regions.

03

Preserve specimen identity

Retain coordinates, expert labels, and crop settings.

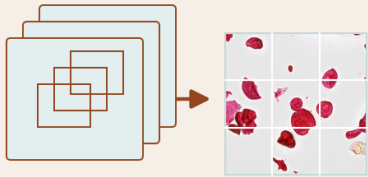
Same source groups, category mapping, and preprocessing for both models.

Joint localization and category identification

A Locate and identify in one model.

PROPOSED ARCHITECTURE / MULTICLASS RF-DETR

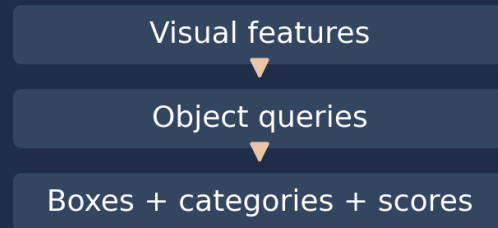
01 Prepare images



Focus stack → tiles

Read small regions

02 Multiclass RF-DETR



Joint localization + identification

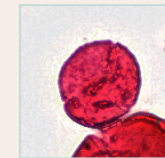
03 Merge predictions



Use slide coordinates

Merge duplicate boxes

04 Expert review



Labeled specimen

Location + score

FAIR COMPARISON

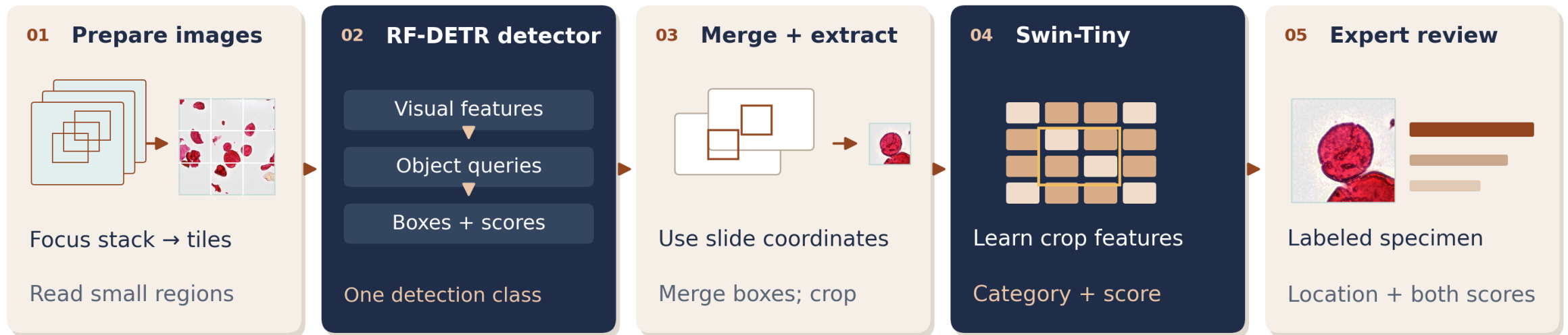
Same category map • same grouped split • same preprocessing • held-out evaluation

Advantage: one model. Challenge: rare or similar categories.

Separate localization from category identification

B Locate first. Identify second.

PROPOSED ARCHITECTURE / RF-DETR + SWIN-TINY



FAIR COMPARISON

Same category map • same grouped split • same preprocessing • held-out evaluation

Advantage: focused classification. Challenge: detection errors propagate.

Test on separate source groups; report errors by category

Detection

Missed specimens
and false detections

Classification

Macro-F1, balanced
accuracy,
and category-level errors

Combined workflow

Specimens both found
and correctly identified

Initial YOLO check: 54/64 and 51/64 known targets recovered

Feasibility only: centered crops and possible prior training exposure.

Compare the models, then make the results useful

01

**Inspect crops
and define
categories**

02

**Freeze evaluation
and train baselines**

03

**Build an expert
review interface**

Deliver specimen images, locations, proposed labels, and uncertainty.